

Matrix Computation I: Norms, Operators, and Numerical Size

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§1.1 Why size matters

Matrix computation starts from a simple question: how large can an error become after a linear operation acts on it? A vector norm measures the size of data; a matrix norm measures the size of an operator. In machine learning language, these are the quantities behind amplification, Lipschitz constants, gradient growth, and numerical stability.

A matrix is an operator

$$x \mapsto Ax.$$

To control it we need both

vector size and operator size.

§1.2 Metrics and completeness

A metric on a set X is a function $d : X \times X \rightarrow [0, \infty)$ satisfying positivity, symmetry, and the triangle inequality. It defines convergence. A sequence is Cauchy if

$$\forall \varepsilon > 0, \exists N, \quad m, n > N \Rightarrow d(x_m, x_n) < \varepsilon.$$

A metric space is complete if every Cauchy sequence converges in it. Completeness means that limiting procedures stay inside the space.

The discrete metric is a useful warning: not every metric expresses the geometry we want. For numerical work, metrics induced by norms are the central examples.

§1.3 Normed vector spaces

A norm satisfies

$$\|x\| = 0 \Leftrightarrow x = 0, \quad \|\alpha x\| = |\alpha| \|x\|, \quad \|x + y\| \leq \|x\| + \|y\|.$$

It induces the metric

$$d(x, y) = \|x - y\|.$$

A complete normed vector space is a Banach space.

In finite dimension, all norms are equivalent: for two norms $\|\cdot\|_a, \|\cdot\|_b$, there exist constants $c, C > 0$ such that

$$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a.$$

Thus finite-dimensional convergence does not depend on the chosen norm, although numerical constants do.

§1.4 Standard vector norms

For $x = (x_1, \dots, x_n)^T$,

$$\|x\|_1 = \sum_i |x_i|, \quad \|x\|_2 = \left(\sum_i |x_i|^2 \right)^{1/2}, \quad \|x\|_\infty = \max_i |x_i|.$$

More generally,

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}, \quad 1 \leq p < \infty.$$

As $p \rightarrow \infty$,

$$\|x\|_p \rightarrow \|x\|_\infty.$$

Holder's inequality gives

$$\sum_i |x_i y_i| \leq \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

and Minkowski's inequality gives the triangle inequality for p -norms.

§1.5 Matrix norms

A matrix norm satisfies the norm axioms and, for computation, should also be submultiplicative:

$$\|AB\| \leq \|A\| \|B\|.$$

This is what allows repeated operations to be bounded:

$$\|A^k\| \leq \|A\|^k.$$

Entrywise examples include

$$m_1(A) = \sum_{i,j} |a_{ij}|, \quad \|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2}.$$

The Frobenius norm also satisfies

$$\|A\|_F^2 = \text{tr}(A^H A).$$

§1.6 Unitary invariance

A matrix U is unitary if $U^H U = I$. It preserves Euclidean length:

$$\|Ux\|_2 = \|x\|_2.$$

The Frobenius norm is unitarily invariant:

$$\|UAV\|_F = \|A\|_F$$

for unitary U, V . This follows from cyclic invariance of trace:

$$\|UAV\|_F^2 = \text{tr}(V^H A^H AV) = \text{tr}(A^H A).$$

Unitary transformations are changes of orthonormal coordinates, not changes of size.

§1.7 Induced norms

Given vector norms on domain and codomain, the induced matrix norm is

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|.$$

It is the maximal amplification factor of A . The standard formulas are

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \quad \|A\|_\infty = \max_i \sum_j |a_{ij}|,$$

and

$$\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^H A)}.$$

The 2-norm is the spectral norm.

§1.8 Normal matrices

A matrix is Hermitian if $A^H = A$, skew-Hermitian if $A^H = -A$, and normal if

$$A^H A = A A^H.$$

Normal matrices are unitarily diagonalizable. If

$$A = U \Lambda U^H,$$

then

$$\|A\|_2 = \max_i |\lambda_i|.$$

For nonnormal matrices, eigenvalues may seriously underestimate transient amplification.

§1.9 Compatibility

A vector norm and a matrix norm are compatible if

$$\|Ax\| \leq \|A\| \|x\|.$$

Every induced matrix norm is compatible with the vector norm that induces it. Compatibility is the bridge from operator size to error propagation:

$$\|A(x + \delta x) - Ax\| \leq \|A\| \|\delta x\|.$$

§1.10 Worked constrained maxima

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Then A is Hermitian with eigenvalues 2, 2, 0. Hence

$$\|A\|_2 = 2.$$

The column sums are all 2, so $\|A\|_1 = 2$. Therefore

$$\max_{\|x\|_1=4} \|Ax\|_1 = 8.$$

If U is unitary and $\|Ux\|_2 = 3$, then $\|x\|_2 = 3$, hence

$$\max_{\|Ux\|_2=3} \|Ax\|_2 = 6.$$

The method is: translate the constraint into a norm ball and apply the induced norm.

§1.11 ML-facing interpretation

A linear layer has Lipschitz constant at most $\|W\|_2$ in Euclidean norm. A stack obeys

$$\|W_L \cdots W_1\|_2 \leq \prod_{\ell=1}^L \|W_\ell\|_2.$$

This inequality is crude but important: moderate layerwise amplification can accumulate over depth. Spectral normalization, orthogonal initialization, residual scaling, and weight decay are all connected to the same numerical question: how large can a perturbation become?

§1.12 Checklist

metric	closeness
norm	vector size
matrix norm	matrix size
induced norm	maximal amplification
Frobenius norm	entrywise Euclidean size
spectral norm	largest singular value